NETWORK SERVICES · DS REVISION

Network DS Cheat Sheet

Chapter 1 (IPv4 & Subnetting) + Chapter 2 (Switching & VLANs).
Every exercise from the sheet is solved step by step, with variants, plus all the course theory that is not in the exercises.

🎯 What the DS will probably look like

VERY LIKELY · EXERCISE 1

Equal subnets (FLSM)

One network → N equal subnets. Bits to borrow, new mask, hosts, the subnet table, "which subnet contains X?", "can X be given to a PC?". → **page 3**

VERY LIKELY · EXERCISE 2

Variable subnets (VLSM)

Departments with different host needs. Sort, host bits, prefix, mask, allocate largest first, table, unused space, "why VLSM?". → **page 5**

MAYBE · EXERCISE 3

VLANs + trunk (Cisco)

Create VLANs, access ports, 802.1Q trunk, allowed VLANs, show commands, ping results and the "why" explanation. → **page 7**

COURSE QUESTIONS

Theory not in the exercises

OSI/TCP-IP, classes, private IPs, ping/tracert, CIDR & summarization, switch MAC table, domains, VLAN types, native VLAN, DTP, attacks, troubleshooting. → **page 10+**

🗺️ Contents

#	Part	What you get
1	**The Toolkit**	The one table that solves 90% of subnetting (prefix → mask → block → hosts), binary in 10 seconds
2	**Exercise 1 · FLSM**	6-step method, full solution, 3 variants (8 subnets, /16 network, by hosts)
3	**Exercise 2 · VLSM**	Method, full solution, ready-to-write explanations, 1 variant
4	**Exercise 3 · VLANs**	Topology, full S1 & S2 config, expected outputs, ping table, explanation, extra commands
5	**Chapter 1 theory**	OSI, encapsulation, addresses, devices, TCP/UDP, ports, IPv4, classes, private, ping, tracert, CIDR, summarization
6	**Chapter 2 theory**	LAN design, switch logic, domains, VLAN types, 802.1Q, native VLAN, DTP, attacks, troubleshooting
7	**Mock DS drills**	Practice questions in the teacher's style, answers at the end
8	**Last-minute checklist**	The traps teachers love

⏱️ How to use this in the last hours

Learn the **Toolkit table** by heart (page 2). Then redo Exercise 1 and Exercise 2 **on paper without looking**, and check against the solutions. Then read the VLAN config twice and write it once from memory. Finish with the theory pages and the checklist.

1 The Toolkit · learn this page by heart

used in Ex 1 & 2

An IPv4 address = 32 bits = 4 octets = network part + host part

192.168.10.150 /26

192	168	10	10	010110
network part (from the original /24)			borrowed	host bits (h = 6)

prefix /26 = 26 bits at 1 in the mask → 255.255.255.192 — 32 − 26 = 6 zeros

🔢 The 3 formulas

Subnets = 2^n (n = borrowed bits)

Usable hosts = $2^h - 2$ (h = host bits left)

Block size = 256 − (interesting mask octet) = 2^h (in the last octet)

"−2" because the first address = **network**, the last = **broadcast**. Neither can go to a PC.

🧠 Powers of 2 (just write them on your draft)

n	1	2	3	4	5	6	7	8
2^n	2	4	8	16	32	64	128	256
2^n-2	0	2	6	14	30	62	126	254

⭐ THE magic table (last octet) · prefix → mask → block → hosts

Prefix	Mask (last octet)	Mask bits (last octet)	Borrowed from /24	Subnets of a /24	Block size	Usable hosts	Network addresses (last octet)
/24	255.255.255.**0**	`00000000`	0	1	256	254	0
/25	255.255.255.**128**	`10000000`	1	2	128	126	0, 128
/26	255.255.255.**192**	`11000000`	2	4	64	62	0, 64, 128, 192
/27	255.255.255.**224**	`11100000`	3	8	32	30	0, 32, 64, 96, 128 …
/28	255.255.255.**240**	`11110000`	4	16	16	14	0, 16, 32, 48 …
/29	255.255.255.**248**	`11111000`	5	32	8	6	0, 8, 16, 24 …
/30	255.255.255.**252**	`11111100`	6	64	4	2	0, 4, 8 … (router links)

Same values work in the 3rd octet: /16=0, /17=128, /18=192, /19=224, /20=240, /21=248, /22=252, /23=254 (block size counts in the 3rd octet, hosts = 2^h-2 with h = 32 − prefix).

⚡ Decimal ↔ binary in 10 seconds

128	64	32	16	8	4	2	1
1	0	0	1	0	1	1	0

150 → take what fits from the left: 128 ✔ (22 left), 64 ✘, 32 ✘, 16 ✔ (6), 8 ✘, 4 ✔ (2), 2 ✔ (0), 1 ✘ → `10010110`

11110101 → 128+64+32+16+4+1 = **245**

🧮 Find the network of ANY address (no binary needed)

1. Block = 256 − mask octet (e.g. /27 → 256 − 224 = **32**)
2. Largest multiple of the block ≤ the address octet: 77 → **64**
3. Network = .64 · Broadcast = next network − 1 = **.95**
4. First host = network + 1 (.65) · Last = broadcast − 1 (.94)

This is exactly **IP AND mask**, done the fast way.

📍 Network address

All host bits = **0**. Names the subnet. Not for a PC.

📣 Broadcast address

All host bits = **1**. To everybody in the subnet. Not for a PC.

💻 Host addresses

Everything between them. The **first usable** usually = router / default gateway.

2 Exercise 1 · Equal-size subnets (FLSM)

will come again, changed a little

Statement: 192.168.10.0/24 → divide into **4 equal subnets** for four departments.

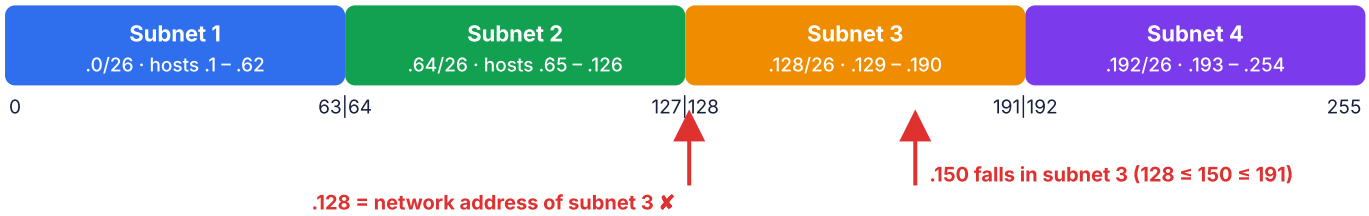
🛠️ The 6-step method (works for ANY version of this exercise)

1. **Borrowed bits n**: smallest n with $2^n \geq$ number of subnets asked. (4 → n = 2; 5 or 6 or 7 or 8 → n = 3)
2. **New prefix** = old prefix + n → /24 + 2 = **/26**. Mask from the magic table → **255.255.255.192**
3. **Subnets created** = 2^n = **4**
4. **Host bits** h = 32 − new prefix = 6 → **usable hosts** = $2^6 - 2$ = **62**
5. **Block size** = 256 − 192 = **64** → networks go 0, 64, 128, 192
6. For each block: network | network+1 | next network − 2 | next network − 1

✅ Full solution

Q	Answer
1. Bits borrowed	`2 bits (2² = 4 subnets)`
2. New mask	`CIDR: /26 · Dotted: 255.255.255.192`
3. Subnets	`2² = 4 subnets`
4. Hosts / subnet	`h = 32 - 26 = 6 → 2⁶ - 2 = 62 usable hosts`

The /24 cut into 4 blocks of 64 (last octet shown)



Subnet	Network address	First usable IP	Last usable IP	Broadcast
1	`192.168.10.0/26`	`192.168.10.1`	`192.168.10.62`	`192.168.10.63`
2	`192.168.10.64/26`	`192.168.10.65`	`192.168.10.126`	`192.168.10.127`
3	`192.168.10.128/26`	`192.168.10.129`	`192.168.10.190`	`192.168.10.191`
4	`192.168.10.192/26`	`192.168.10.193`	`192.168.10.254`	`192.168.10.255`

Q6. Which subnet contains 192.168.10.150?
Block = 64. 150 is between 128 and 191 → **Subnet 3 (192.168.10.128/26)**. (Proof with AND: 150 = 10010110, mask 192 = 11000000 → 10000000 = 128.)

Q7. Can 192.168.10.128 be assigned to a PC?
No. With /26, 192.168.10.128 is the **network address** of subnet 3 (all 6 host bits = 0: `10|000000`). A network address identifies the subnet and cannot be given to a host. The first address a PC can use is 192.168.10.129.
(Same logic for .127, .191, .255 → broadcast addresses.)

⚠️ Careful: with the original /24, .128 would be a normal host! Always answer **with the new mask**.

2b Exercise 1 · How the teacher can change it

train on these

Variant A · 192.168.5.0/24 into 8 subnets (or "6 departments" → still 8!)

$2^3 = 8 \geq 6$ → borrow **3 bits** → **/27** → 255.255.255.**224** → block **32** → 2^5-2 = **30 hosts**. If they asked 6 departments, 8 subnets are created and 2 stay free.

#	Network	First	Last	Broadcast	#	Network	First	Last	Broadcast
1	.0	.1	.30	.31	5	.128	.129	.158	.159
2	.32	.33	.62	.63	6	.160	.161	.190	.191
3	.64	.65	.94	.95	7	.192	.193	.222	.223
4	.96	.97	.126	.127	8	.224	.225	.254	.255

Contains 192.168.5.100? → block 32: 96 ≤ 100 ≤ 127 → subnet 4. Can .160 go to a PC? No, network of subnet 6. Can .95? No, broadcast of subnet 3.

Variant B · 2 subnets: 192.168.1.0/24 → borrow 1 bit → /25 (255.255.255.128), 126 hosts each

#	Network	First	Last	Broadcast
1	192.168.1.0/25	192.168.1.1	192.168.1.126	192.168.1.127
2	192.168.1.128/25	192.168.1.129	192.168.1.254	192.168.1.255

Variant C · A class B network: 172.16.0.0/16 into 4 subnets (the block moves to the 3rd octet!)

Borrow 2 bits → **/18** → mask **255.255.192.0** → block 64 **in the 3rd octet** → h = 14 → $2^{14} - 2$ = **16 382 hosts**.

#	Network	First usable	Last usable	Broadcast
1	172.16.0.0/18	172.16.0.1	172.16.63.254	172.16.63.255
2	172.16.64.0/18	172.16.64.1	172.16.127.254	172.16.127.255
3	172.16.128.0/18	172.16.128.1	172.16.191.254	172.16.191.255
4	172.16.192.0/18	172.16.192.1	172.16.255.254	172.16.255.255

Rule: the last octet of a network address is .0 and of a broadcast .255 when the block is in the 3rd octet.

Variant D · They give **hosts per subnet** instead of number of subnets

"192.168.10.0/24, each subnet needs **25 hosts**. How many subnets?" → smallest h with $2^h - 2 \geq 25$ → h = 5 (30 hosts) → prefix = 32 − 5 = **/27** → borrowed = 27 − 24 = 3 → **8 subnets**.

Then build the table exactly like Variant A.

💡 **Self-check before you hand in:** (1) every broadcast = next network − 1; (2) last broadcast must be **.255**; (3) block × number of subnets = 256.

3 Exercise 2 · VLSM (different sizes)

will come again, changed a little

Statement: 192.168.20.0/24 · Administration **100** · Computer Science **50** · Research Lab **25** · Guest **10** hosts.

🛠️ The VLSM method

1. **Sort** from biggest to smallest number of hosts.
2. For each: smallest **h** with $2^h - 2 \geq$ hosts needed.
3. **Prefix** = 32 − h → mask from the magic table. Block = 2^h.
4. **Allocate** starting at the given network: biggest first. Next network = current network + block.
5. Fill first / last / broadcast like in Exercise 1. Check nothing overlaps.

Q1. Why VLSM instead of the same mask everywhere?
With one mask for everyone, every subnet must be as big as the biggest need (100 hosts → /25). A /24 only has **two** /25 subnets, so we cannot make 4 subnets. With 4 equal subnets (/26, 62 hosts) Administration (100) does not fit. **VLSM** gives each department a mask adapted to its size, so all 4 fit and fewer addresses are wasted.

Q2. Sorted: Administration (100) → Computer Science (50) → Research Lab (25) → Guest (10)

Q3 · Department	Hosts	Min host bits	Check 2^h-2	Prefix	Mask	Block
Administration	100	7	126 ≥ 100 ✔	/25	255.255.255.128	128
Computer Science	50	6	62 ≥ 50 ✔	/26	255.255.255.192	64
Research Lab	25	5	30 ≥ 25 ✔	/27	255.255.255.224	32
Guest Network	10	4	14 ≥ 10 ✔	/28	255.255.255.240	16

Q4 · Allocation, biggest first (bar = the 256 addresses of 192.168.20.0/24)



next network = previous network + previous block size → 0 + 128 = 128 → 128 + 64 = 192 → 192 + 32 = 224 → 224 + 16 = 240

Q5 · Department	Network	Prefix	First usable	Last usable	Broadcast
Administration	192.168.20.0	/25	192.168.20.1	192.168.20.126	192.168.20.127
Computer Science	192.168.20.128	/26	192.168.20.129	192.168.20.190	192.168.20.191
Research Lab	192.168.20.192	/27	192.168.20.193	192.168.20.222	192.168.20.223
Guest Network	192.168.20.224	/28	192.168.20.225	192.168.20.238	192.168.20.239

Q6. Remaining unused space
Used: 128 + 64 + 32 + 16 = **240** addresses. Remaining: 256 − 240 = **16 addresses** → **192.168.20.240/28** (192.168.20.240 → 192.168.20.255), kept for future use (e.g. a /28 or four /30 router links). Bonus: inside the subnets, 26 + 12 + 5 + 4 = 47 usable addresses are not used by hosts.

Q7. One advantage of VLSM in enterprise design
Efficient use of addresses: each subnet gets exactly the size it needs, so less waste and free space stays available for growth. Other valid answers: allows route summarization (smaller routing tables); allows tiny /30 subnets for router-to-router links.

3b Exercise 2 · Variant + traps

train on this

Variant · 192.168.30.0/24: Sales 60 · HR 28 · IT 12 · Link R1-R2 2 hosts

Dept	h	Prefix / mask	Network	First	Last	Broadcast
Sales (60)	6 (62)	/26 · .192	192.168.30.0	.1	.62	.63
HR (28)	5 (30)	/27 · .224	192.168.30.64	.65	.94	.95
IT (12)	4 (14)	/28 · .240	192.168.30.96	.97	.110	.111
Link (2)	2 (2)	/30 · .252	192.168.30.112	.113	.114	.115

Used 64+32+16+4 = 116 → unused 256 − 116 = **140 addresses** (192.168.30.116 → .255).

Trap 1 · exact limits. 62 hosts → /26 works (62 ≥ 62). But **64 hosts → /25** (62 is not enough!). 30 → /27, 31 → /26. 126 → /25, 127 → /24.

Trap 2 · order. Always allocate the **biggest first**. If you start with the small ones, the big blocks won't start on a valid boundary (a /25 must start at .0 or .128).

Trap 3 · gateway. If the statement says "hosts + the router interface", add 1 to the host count before choosing h.

Trap 4 · overlap. Next network = previous network + previous **block** (not + hosts!). Broadcast is never the same as the next network.

📋 FLSM vs VLSM in one look

	FLSM (Exercise 1)	VLSM (Exercise 2)
Mask	Same for every subnet	Different for each subnet
Start from	Number of subnets → borrowed bits	Hosts of each department → host bits
Waste	High if sizes are very different	Low
Routing	Works with classful protocols (RIPv1)	Needs classless protocols (mask sent in updates)
Definition	VLSM = "subnetting a subnet" · used by CIDR to save IPv4 addresses	

4 Exercise 3 · VLANs + 802.1Q trunk (Cisco 2960)

maybe in the DS



Access ports (one VLAN, untagged) → PCs · Trunk port (many VLANs, tagged) → between switches

Q1–Q4 · Configuration of S1

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
! Q1 - create + name the VLANs
S1(config)# vlan 10
S1(config-vlan)# name ADMIN
S1(config-vlan)# vlan 20
S1(config-vlan)# name ENGINEERING
S1(config-vlan)# exit
! Q2 - access ports
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
! Q3 + Q4 - trunk 802.1Q, only 10 and 20
S1(config-if)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20
S1(config-if)# end
S1# copy running-config startup-config
```

Configuration of S2 (identical!)

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S2
S2(config)# vlan 10
S2(config-vlan)# name ADMIN
S2(config-vlan)# vlan 20
S2(config-vlan)# name ENGINEERING
S2(config-vlan)# exit
S2(config)# interface fa0/1
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
S2(config-if)# interface fa0/2
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 20
S2(config-if)# interface g0/1
S2(config-if)# switchport mode trunk
S2(config-if)# switchport trunk allowed vlan 10,20
S2(config-if)# end
S2# copy running-config startup-config
```

"Using IEEE 802.1Q": the 2960 supports only 802.1Q, so `switchport mode trunk` is enough. On a 3560 type first `switchport trunk encapsulation dot1q`. Write a sentence saying so in the DS. Use the port really connected (G0/1, or Fa0/24 if the statement says so).

4b Exercise 3 · Verify, test, explain

Q5 · show vlan brief (on S1)

```
S1# show vlan brief
VLAN Name              Status    Ports
---- ----------------- --------- -------------------------------
1    default           active    Fa0/3 ... Fa0/24, Gi0/2
10   ADMIN             active    Fa0/1
20   ENGINEERING       active    Fa0/2
1002-1005 (default legacy VLANs) ...
```

✔ VLANs exist with the right names ✔ Fa0/1 in 10, Fa0/2 in 20 ✔ **Gi0/1 is NOT listed: trunk ports never appear in show vlan brief.**

Q5 · show interfaces trunk (on S1)

```
S1# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gi0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/1       10,20

Port        Vlans allowed and active in management domain
Gi0/1       10,20
```

✔ Status **trunking** ✔ encapsulation **802.1q** ✔ allowed = **10,20** ✔ native VLAN identical on both sides. Run it on S2 too and compare.

Q6 · Ping tests

Test	Result	Why
a. PC1 → PC3 (192.168.10.20)	✔ Success	Same VLAN 10 + same subnet 192.168.10.0/24. Frame goes S1 → tagged 10 on the trunk → S2 → PC3.
b. PC2 → PC4 (192.168.20.20)	✔ Success	Same VLAN 20 + same subnet 192.168.20.0/24, VLAN 20 allowed on the trunk.
c. PC1 → PC2 (192.168.20.10)	✘ Fails	Different VLANs and different subnets, and there is no router / Layer 3 switch.



Q7. Why can PC1 talk to PC3 but not directly to PC2?
PC1 and PC3 are in the **same VLAN (10)** and the **same IP subnet (192.168.10.0/24)**. The trunk carries VLAN 10 between S1 and S2 (802.1Q tag), so they are in the same broadcast domain and communicate at Layer 2. PC2 is in **VLAN 20** and subnet **192.168.20.0/24**. Each VLAN is a separate broadcast domain (1 VLAN = 1 broadcast domain = 1 subnet), and switches never forward frames from one VLAN to another. To talk between VLANs we need a **Layer 3 device** (router / router-on-a-stick or Layer 3 switch) doing **inter-VLAN routing**, and the PCs need a default gateway. There is none here, so the ping fails.

4c Exercise 3 · Extra commands the teacher can add

from Chapter 2

The DS version of Exercise 3 may add a native VLAN, DTP, more ports, voice VLAN or a management IP. Here is each one:

```
! Native VLAN (unused, same on both ends) + disable DTP on the trunk
S1(config)# vlan 999
S1(config-vlan)# name NATIVE_UNUSED
S1(config)# interface g0/1
S1(config-if)# switchport trunk native vlan 999
S1(config-if)# switchport nonegotiate
! Add a VLAN to the trunk WITHOUT erasing the list
S1(config-if)# switchport trunk allowed vlan add 30
! Several ports at once + description + voice VLAN
S1(config)# interface range fa0/11 - 20
S1(config-if-range)# description ENGINEERING users
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config-if-range)# switchport voice vlan 40
! Unused ports: shut down
S1(config)# interface range fa0/21 - 24
S1(config-if-range)# shutdown
! Management SVI (remote access to the switch)
S1(config)# interface vlan 99
S1(config-if)# ip address 172.16.2.34 255.255.255.240
S1(config-if)# no shutdown
S1(config)# ip default-gateway 172.16.2.33
```

5 Chapter 1 theory · fundamentals recap

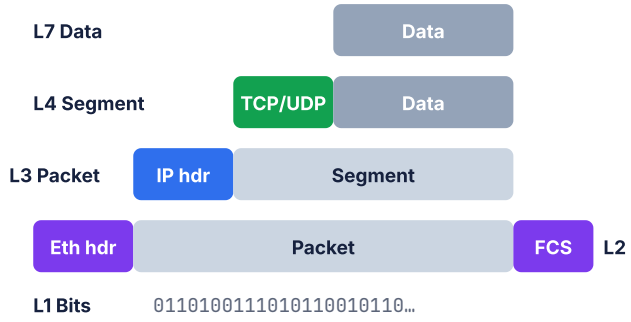
course questions

OSI model (7 layers) — "All People Seem To Need Data Processing"

#	Layer	Role	PDU	Examples
7	Application	Interface with user apps	Data	HTTP, DNS, DHCP, FTP
6	Presentation	Format, encrypt, compress	Data	TLS, JPEG, ASCII
5	Session	Open/manage/close sessions	Data	NetBIOS, RPC
4	Transport	Process-to-process, ports	**Segment**	TCP, UDP
3	Network	Logical addressing, routing	**Packet**	IPv4, IPv6, ICMP
2	Data Link	MAC addressing, medium access	**Frame**	Ethernet, Wi-Fi
1	Physical	Bits on the medium	**Bits**	Cables, hubs

TCP/IP (4 layers): Application (OSI 5-6-7) · Transport (4) · Internet (3) · Network Access (1-2).

Encapsulation



Going down: each layer adds its header (encapsulation).
Going up: removed (decapsulation). IP header = src + dst IP.

The 3 types of addresses

Address	Layer	Size	Example
MAC	2	48 bits (hex)	`00:1A:2B:3C:4D:5E`
IPv4	3	32 bits	`192.168.1.10`
Port	4	16 bits (0–65535)	`80, 443, 53`

> **Key exam idea:** MAC addresses **change at every router** (hop by hop). Source/destination **IP stay the same** end to end.

Devices

Device	Layer	Uses	Collision dom.	Broadcast dom.
Hub	1	nothing (repeats bits)	1 for all	1
Switch	2	MAC table	1 per port	1 (per VLAN)
Router	3	IP routing table	1 per interface	1 per interface

Only a **router** (Layer 3) separates broadcast domains → this is why we subnet. Network = end devices + intermediary devices + media. LAN (building) · WAN (long distance, ISP) · Internet.

TCP vs UDP

	TCP	UDP
Connection	Yes (3-way handshake)	No
Reliability	ACKs + retransmission	Best effort
Ordering	Re-ordered	No
Header	20 bytes, slower	8 bytes, faster
Uses	Web, mail, FTP, SSH	DNS, DHCP, VoIP, video, games

Protocols & ports

Protocol	Port	Transport
HTTP / HTTPS	80 / 443	TCP
DNS	53	UDP (+TCP)
DHCP	67 / 68	UDP
FTP	20 / 21	TCP
SSH / Telnet	22 / 23	TCP
SMTP · POP3 · IMAP	25 · 110 · 143	TCP

DHCP gives 4 things: IP address, subnet mask, default gateway, DNS server.

5b Chapter 1 theory · IPv4, classes, private addresses

course questions

IP protocol = 3 properties

- **Connectionless**: no connection set up before sending.
- **Best effort** (unreliable): no delivery guarantee → TCP adds reliability.
- **Media independent**: works on copper, fiber, wireless.

Network-layer protocols: IPv4, IPv6 (used today) · old: IPX, AppleTalk, CLNS/DECnet.

IPv4 header – key fields

- **Source / Destination address** (32 bits each)
- **TTL** (Time To Live) = hop limit, −1 at each router; 0 → dropped
- **Protocol**: what is inside (TCP, UDP, ICMP...)
- **Flags / Fragment offset**: fragmentation

Classes of addresses (classful)

Class	1st octet	High-order bits	N/H	Default mask	Networks	Hosts / network
A	1 – 127*	`0xxxxxxx`	N.H.H.H	255.0.0.0 /8	128 (2^7)	16 777 214 ($2^{24}-2$)
B	128 – 191	`10xxxxxx`	N.N.H.H	255.255.0.0 /16	16 384 (2^{14})	65 534 ($2^{16}-2$)
C	192 – 223	`110xxxxx`	N.N.N.H	255.255.255.0 /24	2 097 152 (2^{21})	254 (2^8-2)
D	224 – 239	`1110xxxx`	Multicast			
E	240 – 255	`1111xxxx`	Experimental			

* 127.x.x.x = **loopback** (test your own TCP/IP stack). 255.255.255.255 = limited broadcast.

Private addresses (RFC 1918)

Class	Range	Prefix
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8
B	172.**16**.0.0 – 172.**31**.255.255	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16

Used inside a network only, not routed on the Internet (ISPs drop them) → need **NAT** to go out. They help against IPv4 exhaustion. Trap: 172.35.1.1 is **public** (outside 16–31).

Subnet mask & AND

Mask bits = 1 → network part; 0 → host part. Prefix /n = number of 1s.

```
IP    11000000.10101000.00001010.10010110 (…150)
Mask  11111111.11111111.11111111.11000000 (/26)
AND   11000000.10101000.00001010.10000000 = 192.168.10.128
```

AND rule: 1·1 = 1, anything else = 0.

Why subnet? (segmentation) — 3 reasons

- **Performance**: smaller broadcast domains, less useless traffic.
- **Security**: isolate groups, control traffic between them.
- **Manageability**: smaller, organised networks are easier to manage.

+ avoids wasting addresses (IPv4 addresses became insufficient).

IPv4 conservation (history)

- 1981 RFC 791 → classes A, B, C
- **CIDR + VLSM** (1993, RFC 1519)
- **NAT** (1994, RFC 1631)
- **Private addressing** (1996, RFC 1918)

5c Chapter 1 theory · Classful vs classless, CIDR & route summarization

	Classful	Classless
Mask in routing updates	No	Yes
Supports VLSM	No	Yes
Supernet / summary routes	No	Yes
Example	RIPv1	RIPv2, OSPF, EIGRP

Classful protocols don't send the mask because it is deduced from the class. With CIDR the class means nothing, so the mask **must** be sent.

CIDR (Classless Inter-Domain Routing, RFC 1519)

- Introduced by the **IETF** because IPv4 space was running out.
- Uses VLSM ("subnetting a subnet").
- **Advantages:** more efficient use of IPv4 space + **route summarization** (fewer routes in routing tables).

Summary routes use a mask **shorter** than the classful mask.

Route summarization in 3 steps

1. Write the networks in **binary** (only the octet that changes).
2. Count the **left-most matching bits** → that is the new prefix.
3. Copy the matching bits, put **0** everywhere else → summary network.

Course example: 172.20.0.0/16 → 172.23.0.0/16

```
172.20 → 10101100 . 00010100
172.21 → 10101100 . 00010101
172.22 → 10101100 . 00010110
172.23 → 10101100 . 00010111
```

8 + 6 = **14 matching bits** → **172.20.0.0/14** (255.252.0.0)

Lab example: 192.168.0.0 → 192.168.3.0 (/24)

```
0 → 00000000
1 → 00000001
2 → 00000010
3 → 00000011
```

16 + 6 = **22 bits** → **192.168.0.0/22** (255.255.252.0)

Lab example 2: summarize 192.168.10.0/26 and 192.168.10.64/26 (on R2, next hop R1 = 192.168.10.193)

`.0 → 00000000  .64 → 01000000 → 24 + 1 = 25 bits → 192.168.10.0/25 (255.255.255.128)`

```
R2(config)# ip route 192.168.10.0 255.255.255.128 192.168.10.193
! in "show ip route": S = static route, C = directly connected, L = local
```

Course statement: 172.16.0.0/13 summarizes 172.16.0.0/16 to 172.23.0.0/16 (8 networks → 3 bits shorter than /16).

Course examples you must be able to redo

Which IP for PC B? (PC A = 192.168.5.33/27)

/27 → block 32 → network .32, broadcast .63, hosts .33–.62.
A .5 ✘ other subnet · B .32 ✘ network · **C .40 ✔** · D .63 ✘ broadcast · E .75 ✘ other subnet

Which one is a unicast address?

A 224.1.5.2 ✘ multicast (class D) · B FFFF.FFFF.FFFF ✘ broadcast MAC · C 192.168.24.59/30 ✘ broadcast of .56/30 · D 255.255.255.255 ✘ limited broadcast · **E 172.31.128.255/18 ✔** (range 172.31.128.0 – 172.31.191.255, so .128.255 is a normal host)

5d Chapter 1 theory · testing connectivity + warm-up quiz

ping · tracert · ICMP

Testing connectivity: ping (ICMP) — test in this order

1 · ping 127.0.0.1 — local TCP/IP stack → **2 · ping own IP** — NIC configured → **3 · ping gateway** — local network OK → **4 · ping 8.8.8.8** — remote network / routing → **5 · ping google.com** — DNS works

4 works but 5 fails → DNS problem. A failed ping is not always a dead host: a firewall can block ICMP.

ping

Uses **ICMP** (Layer 3): sends **Echo Request**, the host answers **Echo Reply**; measures the round-trip time. TTL in the reply is lower than 64/128/255 because each router subtracted 1. First ping may lose a packet: ARP is resolving the MAC.

traceroute / tracert

Sends packets with TTL = 1, 2, 3… Each router that brings TTL to 0 returns **ICMP Time Exceeded** → the path is discovered **hop by hop**. First hop = your default gateway.

✍️ Warm-up quiz answers (from the course)

1. Router → Layer 3 · 2. Layer-2 PDU → **frame** · 3. At each hop the **MAC** addresses change · 4. DNS → **UDP 53** · 5. A switch does **not** separate broadcast domains (it floods broadcasts on all ports; only routers/VLANs split them) · 6. Application, Transport, Internet, Network Access · 7. The **IP header**, added at **Layer 3** (Network).

Lab 1 questions they can ask

- **ping 8.8.8.8 works but ping www.google.com fails?** → DNS problem (name not translated to an IP).
- **First hop in tracert?** → your default gateway (the router of your LAN).
- **Why is the TTL in the reply lower than 64/128/255?** → each router on the path decreased it by 1.
- **show ip interface brief: "administratively down"?** → the interface was not activated: type `no shutdown`. Working interface = **up / up**.
- **show ip route letters:** C = connected, L = local (the router's own IP), S = static.
- **Which fields change at each hop (simulation mode)?** → source/destination **MAC** and **TTL** change; source/destination **IP** stay the same.
- **Static route syntax:** `ip route <network> <mask> <next-hop>` e.g. `ip route 192.168.10.128 255.255.255.192 192.168.10.194`
- **Cables:** PC↔switch / switch↔router = straight-through; router↔router (same type) = cross-over.

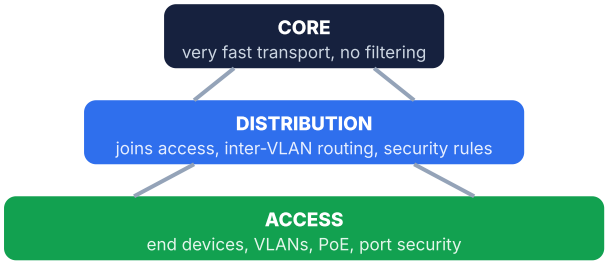
6 Chapter 2 theory · switched networks

course questions

Converged network

Before: one network per service (data, telephone PBX, CCTV video) → 3 cablings, 3 teams, 3 budgets. Today: **one infrastructure for data + voice + video**. Cheaper, but must be well segmented. Switches give power over the cable (**PoE**) and priority to voice (**QoS**).

Hierarchical model



Two links between layers = **redundancy**. **Three-tier** (core + distribution + access): big campus, hospital. **Two-tier / collapsed core** (core+distribution merged): one building, SME, cheaper but limited growth.

Switch: Learn · Flood · Forward · Filter

Action	When
1 Learn	Reads the **source MAC** → writes MAC + incoming port in the MAC table (CAM)
2 Flood	Destination unknown or broadcast (FF:FF...) → all ports except the incoming one
3 Forward	Destination known → only out of its port
4 Filter	Destination is on the same port it came from → drop

Learns from the **source**, decides with the **destination**. Aging time = **300 s** (Cisco). Broadcasts are **always flooded**. Check with `show mac address-table`.

Store-and-forward vs cut-through

	Store-and-forward	Cut-through
How	Whole frame, checks **FCS (CRC)**	Sends after reading **dest MAC**
Errors	Bad frames dropped	Bad frames may pass
Speed	A bit more delay	Lowest latency
Used	Enterprise (Cisco default)	Data centers, HPC, trading

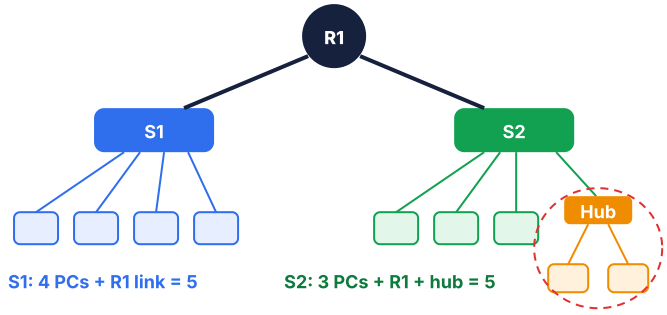
Fragment-free = cut-through that waits for the first **64 bytes** (where collision errors are).

Exercise · MAC table (S1 empty; A on Fa0/1, B Fa0/2, C Fa0/3)

Step	Frame	S1 action	MAC table after
1	A → B	Learn A on Fa0/1; B unknown → **flood** (Fa0/2, Fa0/3)	A-Fa0/1
2	B → A	Learn B on Fa0/2; A known → **forward** to Fa0/1	A-Fa0/1, B-Fa0/2
3	C → FF:FF:FF:FF:FF:FF (ARP)	Learn C on Fa0/3; broadcast → **flood** (Fa0/1, Fa0/2)	A, B, C
4	A → C	A already known (timer reset); C known → **forward** to Fa0/3	A, B, C

6b Chapter 2 theory · domains, VLANs, 802.1Q

Counting collision & broadcast domains



10 collision domains (each switch port = 1; the hub and its PCs share **one**). **2 broadcast domains** (one per router interface).

Rules: Hub = 1 collision domain (half duplex). Switch = 1 per port (full duplex = no collisions). Switch = 1 broadcast domain (per VLAN). Router = 1 broadcast domain per interface. **VLAN = 1 broadcast domain.**

What is a VLAN?

A **Virtual LAN** groups switch ports **logically**, not by place (Engineering can be on 3 floors and still be one network).

1 VLAN = 1 broadcast domain = 1 IP subnet

Devices in different VLANs cannot talk without a **router or Layer 3 switch**.

Why companies use VLANs (6 benefits)

- **Security** – separate PLCs/cameras from guests and offices
- **Performance** – smaller broadcast domains
- **Lower cost** – one switch for several departments
- **Simpler management** – move a user = change a port setting
- **QoS** – voice gets its own VLAN + priority
- **Compliance** – isolate payment (PCI DSS) / patient systems

Types of VLANs

Type	Carries	Example	Good practice
Data	Normal user traffic	10 ADMIN, 20 ENGINEERING	One per department / function
Default	All ports at first boot	VLAN 1	No users in VLAN 1
Native	Untagged traffic on a trunk	999 (unused)	Change it from 1, same on both ends
Management	Access to switches (SSH, SNMP)	99 MGMT	Separate from users, restricted
Voice	IP phones, priority (QoS)	40 VOICE	Separate from data VLAN
Special purpose	OT, cameras, guests	30 PRODUCTION, 60 CCTV, 50 GUEST	Strict isolation

Voice VLAN (one cable, 2 VLANs)

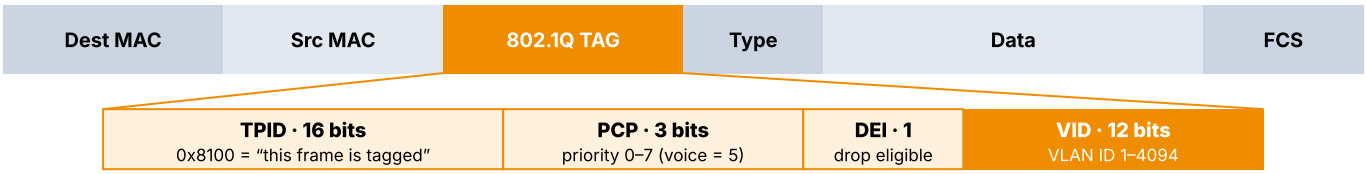
PC plugged into the IP phone, phone into the switch. Phone **tags** voice with VLAN 40 + priority **CoS 5**; PC frames stay **untagged** in VLAN 20. Switch tells the phone the voice VLAN via **CDP/LLDP**. → calls stay clear even during a big download.

VLAN ID ranges

0 and 4095 reserved · **Normal 1–1005** (1002–1005 old Token Ring/FDDI) stored in **vlan.dat** (flash) · **Extended 1006–4094** stored in running-config (providers, data centers). Why 4094? VID = **12 bits** → $2^{12} = 4096 - 2$.

6c Chapter 2 theory · trunks, native VLAN, DTP, security

The 802.1Q tag (4 bytes inserted after the source MAC)



Tag added when the frame enters the trunk, removed before an access port (PCs never see it). Switch recalculates the FCS.

Port	Carries	Connects
Access	1 VLAN, untagged	PC, printer, camera, PLC
Trunk	Many VLANs, tagged (802.1Q)	switch↔switch, router, AP, server

Without trunk: 1 cable per VLAN (20 VLANs = 20 cables) → doesn't scale. If a VLAN is **not allowed** on the trunk or **doesn't exist** on the other switch → frame dropped (most common lab error).

Native VLAN

- On a trunk, frames **without tag** belong to the native VLAN (default VLAN 1).
- Both ends must have the **same** native VLAN, else traffic leaks → log `%CDP-4-NATIVE_VLAN_MISMATCH`.
- Best practice: unused VLAN (999), no ports in it → blocks double tagging.

DTP (Dynamic Trunking Protocol)

S1 \ S2	auto	desirable	trunk	access
dyn. auto	Access	Trunk	Trunk	Access
dyn. desirable	Trunk	Trunk	Trunk	Access
trunk	Trunk	Trunk	Trunk	Limited
access	Access	Access	Limited	Access

Cisco negotiation. Dangerous: an attacker's PC can become a trunk. Rule: fix every port (`mode access` or `mode trunk` + `nonegotiate`). Both ends **auto** = no trunk!

VLAN hopping attacks

Attack	How	Protection
Switch spoofing	PC speaks DTP → gets a trunk → all VLANs	mode access + nonegotiate
Double tagging	Attacker in native VLAN sends 2 tags; 1st switch removes outer one	Native = unused VLAN

VLANs are **not a firewall**: sensitive zones also need filtering rules.

✅ Best-practices checklist

- ☐ No users in VLAN 1
- ☐ Native VLAN = unused (999)
- ☐ Separate management VLAN, SSH only (never Telnet)
- ☐ Unused ports: shutdown + parking VLAN
- ☐ Disable DTP (nonegotiate)
- ☐ Trunks: only needed VLANs
- ☐ Description on every port + VLAN names
- ☐ Document the VLAN plan (IDs, names, subnets)

Cisco ↔ Huawei

Task	Cisco IOS	Huawei VRP
Access port	`switchport mode access`	`port link-type access`
Port in VLAN	`switchport access vlan 10`	`port default vlan 10`
Trunk	`switchport mode trunk`	`port link-type trunk`
Allowed	`…allowed vlan 10,20`	`port trunk allow-pass vlan 10 20`
Show	`show vlan brief`	`display vlan`
Save	`copy run start`	`save`

6d Chapter 2 · CLI modes, troubleshooting, VLSM plan

IOS modes

Prompt	Mode	Enter with
`S1>`	User EXEC	login
`S1#`	Privileged EXEC	`enable`
`S1(config)#`	Global config	`configure terminal`
`S1(config-if)#`	Interface config	`interface fa0/1`
`S1(config-vlan)#`	VLAN config	`vlan 10`

Workflow: 1 create VLANs → 2 access ports → 3 trunks → 4 management SVI → 5 verify & save. `no vlan 10` deletes it → its ports become **inactive**. A port is in VLAN 1 until assigned.

Verification commands

`show vlan brief`	VLANs + access ports (no trunks)
`show interfaces trunk`	trunks, native, allowed list
`show interfaces fa0/1 switchport`	one port in detail
`show mac address-table`	learned MACs
`show running-config`	full config
`copy running-config startup-config`	save

Troubleshooting table

Symptom	Cause	Fix
Port inactive, no network	VLAN deleted / not created	Create the VLAN again
Same VLAN, different switches, no ping	VLAN not allowed on trunk	`allowed vlan add X`
NATIVE_VLAN_MISMATCH log	Native VLAN differs	Same native both ends
Link up but no trunk	Both dynamic auto / access-trunk mismatch	mode trunk both sides
Right VLAN, still no ping	Wrong IP subnet / mask	Correct IP / mask

Work **bottom-up**: cable/port up? → right VLAN? → trunk carries it? → right IP subnet?

Course exercise 5 (trunk troubleshooting)

S1 trunk: native 999, allowed 10,40,99 · S2: native 1, allowed 1-4094. PC1 & PC2 in VLAN 20 can't ping.
Problem 1: VLAN 20 not allowed on S1 → `switchport trunk allowed vlan add 20`
Problem 2: native mismatch 999 vs 1 → on S2 `switchport trunk native vlan 999`
Without **add**, the list would become only 20 and break 10, 40, 99!

Course exercise 3: SahelTech VLAN plan with VLSM (172.16.0.0/16)

VLAN · Name	Hosts	Subnet	Usable	Gateway (1st usable)
50 GUEST	200	172.16.0.0/24	254	172.16.0.1
40 VOICE	120	172.16.1.0/25	126	172.16.1.1
10 ADMIN	50	172.16.1.128/26	62	172.16.1.129
20 ENGINEERING	28	172.16.1.192/27	30	172.16.1.193
60 CCTV	25	172.16.1.224/27	30	172.16.1.225
30 PRODUCTION	20	172.16.2.0/27	30	172.16.2.1
99 MGMT	10	172.16.2.32/28	14	172.16.2.33
999 NATIVE_UNUSED	0	—	—	—

This mixes Ex 2 (VLSM) and Ex 3 (VLANs) — a perfect DS-style combo. Notice how after .224/27 the next block jumps to 172.16.**2**.0.

7 Mock DS drills · do them on paper first

answers next page

Drill 1 · FLSM (like Exercise 1)

The network **192.168.40.0/24** must be divided into **8 equal subnets**.
a) Bits borrowed? b) Mask CIDR + dotted? c) Hosts per subnet? d) Give subnets 1, 2 and 8 (network, first, last, broadcast). e) Which subnet contains 192.168.40.170? f) Can 192.168.40.95 be assigned to a PC?

Drill 2 · VLSM (like Exercise 2)

Network **10.10.10.0/24**: Sales **120**, Marketing **40**, IT **20**, Printers **5**, WAN link **2**.
Give for each: host bits, prefix, mask, network, first, last, broadcast. Then the unused space.

Drill 3 · Find everything from one address

Address	Mask	Network	First	Last	Broadcast	Hosts
192.168.50.77/27	?	?	?	?	?	?
10.1.1.130/25	?	?	?	?	?	?
192.168.24.59/30	?	?	?	?	?	?
172.16.45.200/20	?	?	?	?	?	?

Drill 4 · Class & type

10.45.3.1 · 172.20.5.9 · 172.35.1.1 · 192.168.100.1 · 200.1.2.3 · 224.0.0.5 · 127.0.0.1

Drill 5 · Binary

Convert: 192 · 172 · 10 · 255.255.255.192
And back: 11100000 · 01000001 · 11000000.10101000.00001010.01000001

Drill 6 · Troubleshooting PCs (LAN A 192.168.10.0/26 gw .1 · LAN B .64/26 gw .65 · LAN C .128/26 gw .129)

PC	IP	Mask	Gateway
PC-X (LAN B)	`192.168.10.64`	`255.255.255.192`	`192.168.10.65`
PC-Y (LAN B)	`192.168.10.70`	`255.255.255.0`	`192.168.10.65`
PC-Z (LAN C)	`192.168.10.135`	`255.255.255.192`	`192.168.10.1`
PC-W (LAN A)	`192.168.10.63`	`255.255.255.192`	`192.168.10.1`

Drill 7 · VLAN config

On S1 create VLAN 30 SALES, put Fa0/5–Fa0/8 in it, and add VLAN 30 to the existing trunk G0/1 (which already carries 10,20) without breaking it.

Drill 8 · Short questions

a) Why does show vlan brief not show G0/1? b) What is the native VLAN? c) Why is the max VLAN 4094? d) Does a switch split broadcast domains? e) What does ping use? f) Summarize 10.1.4.0/24 – 10.1.7.0/24.

7b Mock DS drills · answers

Drill 1

a) 3 bits ($2^3 = 8$) · b) /27 = 255.255.255.224 · c) $2^5 - 2 = 30$ · d) S1: .0 | .1 | .30 | .31 · S2: .32 | .33 | .62 | .63 · S8: .224 | .225 | .254 | .255 · e) block 32: 160 ≤ 170 ≤ 191 → **subnet 6 (192.168.40.160/27)** · f) **No**: .95 is the **broadcast** of subnet 3 (.64/27).

Drill 2 (sorted: Sales, Marketing, IT, Printers, WAN)

Dept	h	Prefix / mask	Network	First	Last	Broadcast
Sales 120	7	/25 · 255.255.255.128	10.10.10.0	.1	.126	.127
Marketing 40	6	/26 · 255.255.255.192	10.10.10.128	.129	.190	.191
IT 20	5	/27 · 255.255.255.224	10.10.10.192	.193	.222	.223
Printers 5	3	/29 · 255.255.255.248	10.10.10.224	.225	.230	.231
WAN 2	2	/30 · 255.255.255.252	10.10.10.232	.233	.234	.235

Used 128+64+32+8+4 = 236 → unused **20 addresses** (10.10.10.236 → 10.10.10.255).

Drill 3

Address	Mask	Network	First	Last	Broadcast	Hosts
192.168.50.77/27	255.255.255.224	.64	.65	.94	.95	30
10.1.1.130/25	255.255.255.128	10.1.1.128	.129	.254	.255	126
192.168.24.59/30	255.255.255.252	.56	.57	.58	**.59**	2
172.16.45.200/20	255.255.240.0	172.16.32.0	172.16.32.1	172.16.47.254	172.16.47.255	4094

/20: block 16 in the 3rd octet → 45 is between 32 and 47. And 192.168.24.59/30 is not a valid host because it is the broadcast.

Drill 4

10.45.3.1 A private · 172.20.5.9 B private · 172.35.1.1 B **public** · 192.168.100.1 C private · 200.1.2.3 C public · 224.0.0.5 D multicast · 127.0.0.1 A loopback

Drill 5

192 = 11000000 · 172 = 10101100 · 10 = 00001010 · 255.255.255.192 = 11111111.11111111.11111111.11000000 · 11100000 = 224 · 01000001 = 65 · … = 192.168.10.65

Drill 6

PC-X: .64 is the **network address** of LAN B → use .66 (any .66–.126 free) · PC-Y: wrong **mask** /24 → 255.255.255.192 · PC-Z: wrong **gateway** (.1 is LAN A) → 192.168.10.129 · PC-W: .63 is the **broadcast** of LAN A → use e.g. .4 (.2–.62).

Drill 7

```
S1(config)# vlan 30
S1(config-vlan)# name SALES
S1(config-vlan)# exit
S1(config)# interface range fa0/5 - 8
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# exit
S1(config)# interface g0/1
S1(config-if)# switchport trunk allowed vlan
add 30
```

Do the same VLAN 30 + trunk on S2, or VLAN 30 can't cross.

Drill 8

a) It's a trunk; show vlan brief lists only access ports · b) VLAN of untagged frames on a trunk (default 1, best 999) · c) VID = 12 bits → 4096 − 2 reserved (0, 4095) · d) No (one broadcast domain per VLAN); routers and VLANs split them · e) ICMP Echo Request / Reply · f) 4–7 = 00000**100**…00000**111** → 8+8+6 = **10.1.4.0/22** (255.255.252.0)

8 Last-minute checklist · traps teachers love

🔥 Subnetting traps

- Hosts = 2^h **− 2**. Never forget the −2.
- Subnets = 2^n (no −2 in modern subnetting).
- 5, 6 or 7 subnets → still borrow **3 bits** (8 subnets).
- Answer "can X be assigned?" with the **new mask**, say if it is network or broadcast.
- Broadcast = next network − 1; last broadcast of a /24 = .255.
- VLSM: **biggest first**, next = network + block.
- 64 hosts does NOT fit in /26 (62).
- /17–/23: the block is in the **3rd octet**.
- 172.16–31 private, 172.32+ public.

🔥 VLAN / switching traps

- Create the VLANs on **both** switches.
- `switchport mode access` BEFORE `switchport access vlan`.
- `allowed vlan 30` **replaces** the list → use `add`.
- Trunk ports don't show in `show vlan brief`.
- Same native VLAN on both ends.
- Different VLANs need a **router / L3 switch** (inter-VLAN routing).
- Hub = 1 collision domain for all its PCs.
- Switch learns from **source** MAC, forwards with **destination**.
- Save: `copy running-config startup-config`.

📝 Phrases to write in explanations (they get the points)

- "Each VLAN is a separate **broadcast domain** and a separate **IP subnet**."
- "Communication between VLANs requires a **Layer 3 device** (router or multilayer switch) performing **inter-VLAN routing**."
- "The trunk uses **IEEE 802.1Q** tagging: a 4-byte tag with a 12-bit VLAN ID identifies the VLAN of each frame."
- "X is the **network address** (all host bits = 0) / **broadcast address** (all host bits = 1), so it cannot be assigned to a host."
- "VLSM allocates a mask adapted to each subnet's size, so address space is used **efficiently** and space remains for growth."
- "Subnetting improves **performance** (smaller broadcast domains), **security** and **manageability**."

You've got this. 💪

Draw the magic table on your draft paper the second the DS starts — then every subnetting question is just reading from it.

Built from: Chapter 1 (IPv4 Addressing & Subnetting), Chapter 2 (Switched Networks & VLANs), Lab 1, Exercises_network — Network Services 2026-2027.